[INCH-POUND]
A-A-59257
May 19, 1998

COMMERCIAL ITEM DESCRIPTION

OSCILLOSCOPE
(Digital Storage, 500 MHz)

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE.

This CID outlines the minimum requirements for a 500 Megahertz (MHz), four-channel oscilloscope with a two Giga-sample per second (GS/sec) sampling rate and digital storage. This commercially-available digital storage oscilloscope will be used on a bench and is capable of being modified with a rackmount kit to be included in a rackmount assembly. Only those manufacturers that meet or surpass the requirements outlined in this CID are qualified.

2. SALIENT CHARACTERISTICS.

2.1. Class 3. The oscilloscope covered by this CID is categorized as Class 3 test equipment in the latest version of MIL-PRF-28800F.

2.2. Safety and Environmental. The oscilloscope shall meet all safety and environmental requirements as specified in MIL-PRF-28800F for Class 3 when used as bench-top or rackmount equipment.

2.3. Electrical Power Sources. The oscilloscope shall operate from nominal commercial, military, and shipboard power sources of 110/120 Volts rms (± 5%), single phase at line frequencies of 50, 60 and 400 Hertz (Hz) and 200/240 Volts rms (±5), single phase at line frequencies of 50 and 60 Hz. The oscilloscope shall operate in accordance with the steady state conditions, transient state conditions, and interruption of power source requirements outlined in MIL-PRF-28800F. Power consumption shall not exceed 400 watts.

2.4. Accessories. Accessories to be supplied by the manufacturer with each deliverable oscilloscope shall include standard accessories such as the appropriate power cord(s), probe(s), accessory pouch, front panel cover, a written tutorial/user reference manual, a written programmer reference manual, and a traceable calibration certificate.

2.4.1. Cover. The oscilloscope enclosure shall have a cover protecting the front panel controls and display in accordance with MIL-PRF-28800F.

2.4.2. Probes. The active or passive probes provided for all four channels will permit an input signal attenuation of 1 and 10 over the full 500 MHz bandwidth and vertical accuracy. The probe attenuation shall be automatically recognized and the scale factor of the oscilloscope adjusted appropriately.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: The Resources and Logistics Services Center, SA-ALC/TILDD, Bldg. 171, Post C-12, 485 Quentin Roosevelt Rd., Kelly AFB, TX 78241-6524.

AMSC N/A FSC 6625

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

2.5. Interface. The oscilloscope shall be capable of interfacing with industry standard data interchange formats for files, formats, storage, communications, plotting, and printing.

2.5.1. ANSI/IEEE-STD-488 Interface Requirements. The oscilloscope's digital interface shall control each device function including all ranges, modes, and levels except for primary power control and other non-measurement functions. Requirements shall be as stated in MIL-PRF-28800F, paragraph 3.4.1.

2.5.2. Output Ports. The oscilloscope shall have RS-232 interface that complies with EIA/TIA 574 talk, talk only, and talk/listen commands and Centronics interface complying with C332-44, Feb. 1977, Rev. A, for printer and/or plotter outputs.

2.5.3. Printer Interface. The oscilloscope shall have the minimum capability of interfacing with Laser and Epson style printers.

2.5.4. File Formats. The oscilloscope shall have, at the least, capability of providing, storing, and transferring data in TIFF and BMP image formats.

2.5.5. Data Format. The oscilloscope shall have the ability to process data with MathCAD™ or similar software.

2.6. Rackmount Option. If specified in the contract or order, the oscilloscope shall be delivered with an attachable rackmount kit which includes the necessary accessories and hardware to interconnect all front and rear panel input and output ports and include any filler panels to provide the preferred height and width if required. Conversion to a rackmount configuration shall be accomplished by use of a kit that converts the enclosure, but will not prevent return to the original configuration.

2.7. Dimensions and Weight. The oscilloscope dimensions shall permit the oscilloscope's inclusion in a standard 19-inch wide rack enclosure with a standard height of less than 6 rack units. The oscilloscope's weight, including the front cover and all accessories (excluding written manuals), shall not exceed 48 pounds.

2.8. Preconditioning Requirement. Each deliverable oscilloscope shall be capable of withstanding the environmental screening effects of burn-in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

2.9. Self-Test. The oscilloscope shall provide an automatic diagnostic self-test, in accordance with MIL-PRF-28800F paragraph 3.3.6, automatically upon start-up and/or when invoked by the operator either manually or remotely.

2.10. Extended Diagnostics. The oscilloscope shall provide extended diagnostic routines to detect internal electrical failures to the next lower level from self-test and shall be invoked by front panel controls.

2.11. Signal Path Compensation. The oscilloscope shall provide internal signal path compensation either automatically on start-up and/or when invoked remotely or by front panel controls.

2.11.1. Calibration and Maintenance Adjustments. The oscilloscope shall have an auto calibration feature which ensures specified DC and timing accuracy. Changes in environmental conditions within the operating range specified for the equipment will not require additional calibration of the oscilloscope, except for internal calibration procedures and signal path compensation, to meet performance requirements.

2.12. Autoset. The oscilloscope shall provide an Autoset function that when activated, will automatically configure the vertical, horizontal, triggering, acquisition, and display controls to provide a triggered display positioned on a screen.

2.13. Temperature Requirements. The oscilloscope shall meet performance and accuracy requirements in Class 4 operating and non-operating environments. Disk operations shall be required over the range of 10 to 40 degrees C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

2.14. Humidity. The oscilloscope shall meet performance and accuracy requirements for disk drive operations in a Class 4 operating environment of 5 to 95% (±5%) relative humidity from 10 to 30 degrees C.; to 75% (±5%) relative humidity at 30 to 40 degrees C.; and 45% (± 5%) relative humidity above 40 degrees C.

2.15. Shock, Functional. The oscilloscope shall meet its performance and accuracy requirements in an operating environment after being subjected to a 30 g half-sine shock test.

2.16. Altitude. The maximum operating altitude with hard disk in use will be 10,000 feet, and the oscilloscope will be functional after being transported at a maximum altitude of 40,000 feet.

3. PERFORMANCE CHARACTERISTICS.

3.1. Warm-up Performance. The equipment shall perform to specifications after a warm-up period not to exceed thirty minutes after initiating power and all calibration and self-check routines have been accomplished.

3.2. Acquisition System. This digitizing oscilloscope shall acquire data on four channels simultaneously and will capture waveforms in a single trigger event up to the full rated 500 MHz bandwidth. Examples of acquisition modes include: sample, average, peak detection, envelope, high resolution, fast frame, or single sequence.

3.2.1. Sample Mode. The oscilloscope shall provide a sample mode in real time and/or repetitive sampling which may be sequential, random, interleaved or any other technique to accomplish a reconstruction of the waveform.

3.2.2. Dead Band or Dead Time Between Segments. The oscilloscope's dead band or dead between waveform captures shall be 50 µs or less.

3.2.3. Bandwidth. The oscilloscope bandwidth specifications of 500 MHz shall apply for all channels when coupled at 50 ohm or 1 Mohm. The oscilloscope bandwidth specification applies with the standard accessory probe installed.

3.2.3.1. DC Coupled Bandwidth. The oscilloscope shall have a DC response, with 50 ohm or 1 Mohm input impedance that shall not be more than -3 dB down from DC to 500 MHz for all sensitivity. The bandwidth response shall not exceed +1 dB from DC to 500 MHz.

3.2.3.2. AC Coupled Bandwidth. The AC coupled lower frequency response at 1 Mohm input impedance shall not be more than -3 dB down at 10 Hz and the upper level frequency response shall not be more than -3 dB down at 500 MHz. The bandwidth response shall not exceed +1 dB from 10 Hz to 500 MHz.

3.2.3.3. Bandwidth Limiting. The oscilloscope shall provide the capability to limit the upper bandwidth of all channels over the ranges 20 to 50 MHz, 80 to250 MHz, and full bandwidth.

3.2.3.4. Step Recovery Settings. The oscilloscope shall recover with ≤ 1% error within 60 µs of a ≤ 2 volt step input with a rise time > 35 ηs.

3.2.4. Sample Rate. The oscilloscope shall have a minimum sample rate for real time sampling of 2 GS/s for one or two channel operation and a minimum of 1 GS/s with three or more channels in operation.

3.2.5. Record Length. The oscilloscope shall be capable of acquiring a minimum of 50,000 samples simultaneously for each channel. Additional record lengths will be selectable as provided.

3.2.6. Acquisition Modes. The oscilloscope shall provide the following acquisition modes:

3.2.6.1. Sample. The oscilloscope shall sample data in accordance with an internal algorithm allowing data to be acquired dependent upon the time/division setting and record length selected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

3.2.6.2. Average. The oscilloscope shall sample data over multiple acquisitions and provide an average composite waveform of the data. The average range shall be selectable from 2 to 10,000 waveforms.

3.2.6.3. Peak Detection. The equipment shall capture pulses of at least 3 ηs in duration for the real time sample rate.

3.2.6.4. Envelope Mode. The oscilloscope in a repetitive sampling mode shall capture maximum and minimum points of a series of waveform and store this data for latter display. The range for this capability shall be from 2 waveforms to 10,000 waveforms.

3.2.6.5. Single. The oscilloscope shall capture a single waveform in accordance with an internal algorithm allowing data to be acquired determined by the time/division setting and record length selected either manually or by a triggered input.

3.2.6.6. Increased Resolution. The oscilloscope shall have a mode which will increase vertical resolution from 8 bits to a minimum of 10 bits of vertical resolution.

3.3. Horizontal or Time Base System. The oscilloscope shall provide the following horizontal timing functions:

3.3.1. Seconds/Division Range. The oscilloscope shall provide a seconds/division range of at least 1 ηs/division to 10 s/division in a 1-2-5 sequence.

3.3.2. Time Base Accuracy. The oscilloscope shall have a time base accuracy of at least ± 25 parts per million (ppm) over the temperature range of +15°C to 35°C.

3.3.3. Interpolator Resolution. The oscilloscope shall have a resolution of at least 10 ρsec.

3.3.4. Horizontal Jitter. The oscilloscope shall have less than 50 ρs of horizontal jitter.

3.3.5. Delay Between Channels. The oscilloscope delay between channels shall not exceed 50 ρs for any channels with equal volts/division and coupling settings

3.4. Vertical System. The oscilloscope shall function as follows:

3.4.1. Input Coupling. The oscilloscope shall provide AC coupling and DC coupling for the 1 M input and DC coupling for the 50 input. A ground reference shall be provided for all input channels. When in the ground reference position, the signal input shall be opened and the input circuitry to the amplifier shall be grounded.

3.4.2. Sensitivity Range. The oscilloscope shall provide a sensitivity range for all channels that is selectable in steps from 2 mV/division to 5 V/division in a 1-2-5 sequence, with a X1 probe or direct input. Indicated sensitivities shall be automatically switched to display the correct scale factor when properly coded attenuator probes are attached.

3.4.3. Offset Range. The oscilloscope's offset range for all channels shall be a minimum of ±4 divisions for all Volts/division settings.

3.4.4. Position Range. The oscilloscope shall provide the capability to position each displayed trace from the center of the display to the top and bottom of the display.

3.4.5. Display Dynamic Range. The oscilloscope shall provide the capability to avoid clipping on varying signal artifacts that exceed the top and bottom display area. This may be accomplished by, but not limited to, auto setup features, display indications, warnings, increased dynamic range. These capabilities may be provided either individually or in combination.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

3.4.6. Channel Isolation. The oscilloscope isolation between the input channels, with identical Volts/division settings, shall be at least 40 dB at 100 MHz and at least 30 dB at 500 MHz.

3.4.7. DC Gain Accuracy. The oscilloscope DC gain accuracy shall be at least ± 1.0% of full scale at 0 volts offset.

3.4.8. Input Impedance. The oscilloscope input impedance shall be selectable between 1 Mohm (±1.0%) and 50 (±1.0%). The VSWR for 50 ohm input shall be 1.5:1 from DC to 500 MHz.

3.4.9. Maximum Input Voltage. The oscilloscope shall withstand the following requirements without damage:

3.4.9.1. One Megohm Input Impedance. The oscilloscope inputs shall withstand, without damage, up to ±250 V (DC plus peak AC), derated at 20 dB/decade above 10 kHz when 1 Mohm input impedance selected.

3.4.9.2. 50 Ohm Input Impedance. The oscilloscope inputs shall withstand, without damage, up to 5 V rms with peaks less than or equal to ±30 volts with 50 ohm input impedance selected. The oscilloscope shall be protected from overload of the internal 50 ohm load resistors by sensing over-voltage conditions and automatically switching to 1 Mohm coupling on the overloaded input.

3.4.10. Vertical Resolution. The oscilloscope shall have a minimum of 8 bits of resolution in real time sampling mode and a minimum of 10 bits resolution in repetitive sampling mode. Vertical resolution with real time sampling shall be at least 2% at 500 MHz.

3.4.11. Vertical Zoom. . The oscilloscope shall have the ability to vertically zoom up to 5X expansion or up to 50X expansion with averaging and accomplish this at a sensitivity of 50 µV/division.

3.5. Triggering System. The oscilloscope shall provide both main and delayed triggering, edge triggering, logic triggering, and pulse triggering capabilities. Edge triggering shall be fully configurable for source, slope, coupling, mode, and hold off.

3.5.1. Triggering System Bandwidth. The oscilloscope shall provide triggering over the full 500 MHz bandwidth.

3.5.2. Time Base Trigger Modes. The scope shall provide time base trigger selectable in main and delay modes.

3.5.2.1. Main Trigger . The oscilloscope shall provide the following main trigger modes.

3.5.2.1.1. Normal Mode. The oscilloscope main time base trigger shall be selectable from a vertical amplifier signal, auxiliary signal, or the power line. A waveform shall be acquired only in the presence of a trigger signal. When no trigger signal occurs, there shall be no acquisition of a waveform. Loss of the trigger shall stop any multiple acquisition sequence in progress and freeze the display, regardless of the selected time/division setting.

3.5.2.1.2. Automatic Mode. The oscilloscope shall initiate acquisition when a trigger occurs or after a certain time period (that varies based on the time/division setting) following the end of the last acquisition, whichever occurs first. A waveform shall be displayed even though the oscilloscope is not triggered. Automatic mode uses a timer that starts after a trigger event occurs. If another trigger event is not detected before the timer runs out, the oscilloscope generates another trigger.

3.5.2.1.3. Single Mode. The oscilloscope shall initiate acquisition and shall run one time or in accordance with the acquisition mode selected following rearming for each selected vertical channel and sweep combination, when triggered as in the normal mode. A front-panel control shall rearm the sweep and illuminate a reset indicator.

3.5.2.2. Delay Trigger. The oscilloscope shall have a delayed trigger mode providing delay by time or delay by events.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

3.5.2.2.1. Delay Jitter. The oscilloscope shall have a delay jitter ≤ 40 ppm of the time base delay setting.

3.5.2.2.2. Time Delay. The oscilloscope shall provide a trigger mode that delays the initiation of the display and timing sequence from 20 ηs to 200 seconds.

3.5.2.2.3. Time Delay Accuracy. The oscilloscope delay time accuracy shall be identical to the main time base accuracy over the temperature range of +15°C to +35°C.

3.5.2.2.4. Delay By Events. The oscilloscope shall provide a trigger mode that delays the initiation of the display and timing sequence from 2 to 9,999,999 events. An event is characterized by a positive or negative transition through the trigger level as defined by the type of coupling.

3.5.2.3. Edge Triggering. The oscilloscope shall provide edge triggering in accordance with the following:

3.5.2.3.1. Trigger Level And Slope. The oscilloscope shall provide a continuously variable trigger level control. Slope controls shall also be provided to permit triggering on any level on the positive and negative slopes of a trigger signal.

3.5.2.3.2. Trigger Level or Trigger Threshold Ranges. The upper and lower voltage limits that the trigger level or threshold can be set to are: ± 4 divisions from screen center for any channel trigger source, ±4 Volts for auxiliary trigger source, and ±250 Volts for line trigger source.

3.5.2.3.3. Input Channel. The oscilloscope trigger signal is derived from any one of the primary inputs or trigger channel. If the trigger source is one of the primary channels selected for display, you can view the trigger source. You can also trigger from an input source not being displayed.

3.5.2.3.4. AC Line Voltage. The oscilloscope trigger signal is derived from the AC line voltage. AC line is available as a source only for the Main Time Base trigger.

3.5.2.3.5. External Trigger. The oscilloscope shall provide an external trigger input.

3.5.2.3.5.1. External Trigger Level Range. The scope shall have a trigger level range greater than ± 500 mV.

3.5.2.3.5.2. External Trigger Level Edge Sensitivity. The oscilloscope shall trigger on a minimum of 400 mV up to 50 MHz and a minimum of 750 mV up to 100 MHz.

3.5.2.3.5.3. External Trigger Level Maximum Input. The oscilloscope shall accept external inputs up to ± 20V(DC + peak AC).

3.5.2.3.5.4. External Trigger Bandwidth. The oscilloscope external bandwidth shall be equal to or exceed the normal triggering bandwidth of the equipment.

3.5.2.4. Edge Type Trigger Coupling. The oscilloscope shall provide the following trigger coupling modes:

3.5.2.4.1. Direct Current Trigger (DC). The oscilloscope shall trigger on both AC and DC signal components when signal levels exceed 0.5 divisions to 50 MHz and 1.5 divisions to 500 MHz.

3.5.2.4.2. Alternating Current Trigger (AC). The oscilloscope shall trigger on the AC signal components above approximately 60 Hz when signal levels exceed 0.5 divisions to 50 MHz and 1.5 divisions to 500 MHz.

3.5.2.4.3. High Frequency Rejection. The oscilloscope shall trigger on signals at 0.5 divisions from DC to approximately 30 kHz.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

3.5.2.4.4. Low Frequency Rejection. The oscilloscope shall trigger when signals exceed 0.5 divisions above 80 kHz to 50 MHz and 1.5 divisions to 500 MHz.

3.5.2.5. Trigger Position. The position of the trigger in the waveform record shall be selectable in 1% increments to allow 0 to 100% of the data to be acquired.

3.5.2.5.1. Trigger Jitter. The oscilloscope trigger jitter shall not exceed 100ρs at full bandwidth and the fastest sweep setting.

3.5.2.6. Exclusion Triggering. The oscilloscope shall trigger on a signal or period that fall outside two limits which are settable from 2.5 ηs to 20 seconds.

3.5.2.7. Pulse Triggering. The oscilloscope shall provide triggering on pulse edges with the minimum characteristics described herein.

3.5.2.7.1. Glitch Triggering. The oscilloscope shall trigger or reject signals on the rising or falling edges of pulses with widths at least over the range of 5 ηs to 1 second.

3.5.2.7.1.1. Glitch Triggering Time Accuracy. The oscilloscope shall have a minimum time accuracy for pulse width triggering for a time range of 5 ηs to 1 μs, of ±(20% of setting + 1.5 ηs) and for a time range > 1 μs ≤ 1 s, accuracy shall be ±0.5%.

3.5.2.7.2. Pulse Width Triggering. The oscilloscope shall trigger on a positive or negative pulse within or not within the selectable time limits of at least 2.5 ηs to 1 second.

3.5.2.7.2.1. Time Accuracy For Pulse Width Triggering. The oscilloscope shall have a minimum time accuracy for pulse width triggering for a time range of 2.5 ηs to 1 μs, of ±(20% of setting + 1.5 ηs) and for a time range > 1 μs ≤ 1 s, accuracy shall be ±0.5% measured at the 50% points.

3.5.2.8. Time-out Triggering. The oscilloscope shall trigger when a pulse fails to complete a transition from a high to low state and return to the original state within the time range of at least 2.5 ηs to 20 seconds.

3.5.2.9. Video Triggering. The oscilloscope shall provide video triggering in PAL, NTSC, SECAM, and non-standard video formats in monochrome or color.

3.5.2.9.1. Video Trigger Selection. The oscilloscope shall trigger on the TV input signal referenced to line or field number or both.

3.5.2.9.2. Video Trigger Sensitivity. The oscilloscope shall trigger on 0.8 divisions or less of composite SYNC signal.

3.5.2.9.3. Video Synchronization. The oscilloscope shall trigger at least on falling horizontal synchronization signals.

3.5.2.10. Non-Standard Video Triggering. The oscilloscope shall provide a triggering capability for non-standard video by providing the ability of defining repetition rate, number of lines, number of fields, and vertical synchronous timing to initiate a trigger action.

3. 6. Measurement System Features. The oscilloscope shall provide the following measurement capabilities:

3.6.1. Automated Measurements. The oscilloscope shall provide automated measurements defined by IEEE-STD-181-1977. The oscilloscope measure function will automatically extract parameters from any signal input to the scope. Four or more of the parameters can be displayed on screen and made available for transfer over the IEEE-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

488 interface bus. The parameters measured shall include, but are not limited to: period, frequency, width, rise time, fall time, duty cycle, delay, positive overshoot, negative overshoot, maximum, minimum, peak-to-peak, amplitude, root mean square (rms), propagation delay, burst, mean and duration.

3.6.2. Automated Measurement Accuracy Voltage. The scope shall have a minimum accuracy of ± 2% for voltage related measurements, for single and dual point measurements with the signal at 50% or higher of full scale.

3.6.3. Automated Measurement Accuracy Time. The oscilloscope shall have minimum accuracy of ± 1% for time related measurements for single and dual point measurements.

3.6.4. Cursor Measurements. The oscilloscope shall have the capability to make single and dual cursor measurements along both the vertical and horizontal axis with and without tracking. Cursors shall be provided for making vertical and horizontal measurements on the displayed waveforms. The cursors shall have the ability to measure absolute and delta voltage between the positions of the horizontal bar cursors and absolute and delta time in seconds or Hz for 1/time between the positions of the vertical bar cursors. The time and voltage related values represented by cursor positions shall be displayed on screen and available for transfer over the IEEE-488 interface bus.

3.6.4.1. Cursor Measurement Accuracy Voltage. The oscilloscope shall have a minimum accuracy of ± 2.0% of full scale for voltage related measurements when utilizing single cursors and ± 4.0% when using dual cursors.

3.6.4.2. Cursor Measurement Accuracy Time. The oscilloscope shall have a minimum accuracy of ± 1.0% of full scale for time related measurements when utilizing either single or dual cursors.

3.6.4.3. Cursor Readout. The oscilloscope shall provide a cursor readout selectable by the user to display cursor measurement in either voltage, time, or frequency or any combination of these measurements. Dual cursor measurements shall provide a read-out of the relative positions of the cursors and the difference.

3.6.5. Math Functions. The oscilloscope shall provide math waveform operations of addition, subtraction, inversion, and multiplication of any two waveforms in any combination from the current displayed waveforms or stored waveforms.

3.6.6. Pass/Fail and Limit Testing. The oscilloscope shall have the capability to define limits and patterns for pass/fail and limit testing.

3.6.6.1. Waveform Failure Indication. The oscilloscope shall have the ability to indicate waveforms out of limits by the following means by use of: 1) waveform parameter limits or with "masks" which describe a waveform's shape, 2) a failure invoking an operator preprogrammed action, or 3) a combination of both 1) and 2).

3.6.6.2. Forms of Pre-Programmed Failure Indications. The oscilloscope shall display failures through its automated waveform measurement capability by: printing a hardcopy report of the failure; saving the failure in memory; responding to a GPIB PC computer-generated SRQ (service request); activating an audible alarm; or any combination of the preceding.

3.7. Display, Controls, & Adjustments. The oscilloscope shall provide the following display, controls and adjustments:

3.7.1. Display. A color display or visual device capable of displaying both waveforms in normal ambient room light in sufficient detail for direct measurements on screen without automated measurements or cursors. The minimum viewing area of the display shall be at least 10 centimeters (cm) horizontally and 8 cm vertically.

3.7.2. Display Style. The oscilloscope shall provide waveform displays in sin (x)/x, linear, dots, variable persistence, and infinite persistence styles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

3.7.3. Trace Convergence. The oscilloscope shall have the capability to discriminate and display where multiple waveforms converge.

3.7.4. Color Conventions. The oscilloscope color display shall exhibit at least seven colors of the visible light spectrum, which are: red, orange, yellow, green, blue, indigo, and violet.

3.7.4.1. Color Grading. The oscilloscope shall have color grading and exhibit such when the variable persistence mode is selected. The waveform points shall time decay through a minimum of 12 levels of intensity.

3.7.4.2. Color Resolution. The oscilloscope shall have the resolution of a standard VGA monitor, or better such that the resolution is 640 horizontal points by 480 vertical points.

3.7.5. Display Intensity. The oscilloscope shall provide the capability to control overall intensity, text/graticule intensity, and waveform intensity.

3.7.6. Display Persistence. The oscilloscope shall provide both infinite and variable persistence. The variable persistence range shall be over the range of 500 ms to 10 seconds with at least 50 ms resolution.

3.7.7. Display Gray Scaling. The oscilloscope shall provide the ability to accomplish gray scaling of at least 16 discernable intensity levels.

3.7.8. Display Graticule Size and Markings. The oscilloscope performance shall be over a minimum 8 X 10 division graticule in the waveform display area. The cardinal axes shall be ruled in 0.2 minor divisions.

3.7.9. Display Graticule Types. The oscilloscope shall provide the capability of selecting full screen, single window, dual window, and XY, grid, or crosshair displays.

3.7.10. Waveform Display Area. The oscilloscope waveform display area shall have at least 640 horizontal points by at least 480 vertical points.

3.7.11. Screen Data Displayed. Horizontal and vertical scaling factors for each channel, main and delayed time bases, trigger sources, trigger level, trigger type, sample rate, mode, at least four measurements, and a representation of memory length with trigger pint indicated.

3.7.12. VGA Compatible Output. The oscilloscope shall have a VGA compatible output allowing an external VGA monitor to be connected to it.

3.7.13. VGA Compatible Display Connector. The oscilloscope shall have a D-type VGA connector with 15 pins.

3.7.14. Format. The oscilloscope shall have both YT and XY axis formats.

3.7.15. "Fit to" and "Fill" The Screen. The oscilloscope shall have the ability to display the entire memory acquisition on the screen.

3.7.16. Enhanced Video Resolution. The scope shall be able to display and analyze at least 1,500 lines per inch of video signals.

3.7.17. Magnification. The oscilloscope shall provide a horizontal magnification adjustable over a minimum range of 1 to 1000. The oscilloscope shall position waveforms both vertically and horizontally and read out on screen of the vertical and horizontal scaling factors.

3.7.18. X-Y Operation. The oscilloscope shall provide X-Y operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

3.7.18.1. X-Axis and Impedance Range. The scope shall have an X-axis deflection factor, variable range, and accuracy as well as an input impedance range defined for channels 1 and 2.

3.7.19. X and Y Phase Difference. The oscilloscope phase difference between X and Y shall be less than 1° from DC to 1 MHz and less than 2° from 1 MHz to 2 MHz.

3.8. Waveform/Setup Storage. The oscilloscope shall provide internal waveform storage to store waveforms, reference waveforms, front panel setups, and calibration data.

3.8.1. Waveform and Set-Up Storage. The oscilloscope shall provide internal memory to store a minimum of four waveforms at maximum memory storage and sufficient internal memory to store four front panel set-ups.

3.8.2. Back Up Battery. The oscilloscope shall have a rechargeable back up battery with an internal memory of at least two years. The equipment shall have the capability to store front panel settings and to retain this data for a minimum of two years. If the equipment uses a nickle-cadium battery, there shall be means by which the operator and/or internal circuitry can deep discharge the battery to prevent "memory effect."

3.8.3. Memory Data Transfer. The oscilloscope shall provide the capability to save any waveform or setup data to any applicable reference memory location, to transfer setup information and/or waveforms into a reference memory location via the GPIB interface, and to transfer setup and waveform data from the floppy disk drive.

3.8.4. Mass Storage. The oscilloscope shall provide mass storage providing additional storage of reference waveforms, setups, data files and image files. The oscilloscope may have a floppy disk in a 3.5 inch size, 1.4 Megabytes storage, and in a DOS format.

3.8.5. Alternate Mass Storage Devices. The oscilloscope may use a PCMCIA I/II/III flash memory card or other storage device with 1.4 Megabytes or greater capacity and DOS format.

3.8.6. Mass Storage Operational Restraints. The oscilloscope shall have the ability store, archive, retrieve, and any other storage operation over a temperature range of +10°C to +40°C.

4. REGULATORY PROVISIONS.

In accordance with Section 23.403 of the Federal Acquisition Regulation, the federal government's policy is to acquire items composed of the highest percentage of recoverable materials practicable consistent with maintaining a satisfactory level of competition and without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

5. QUALITY ASSURANCE PROVISIONS.

5.1. Product Conformance. The oscilloscope provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The federal government reserves the right to require proof of such conformance.

5.2. Submitted Products. The equipment submitted for bid sample testing shall be representative of the commercial off-the-shelf product. All features, performance, and environmental protection of the product tested during bid sample testing shall be provided under any resulting contract or purchase order.

5.3. Calibration. The oscilloscope shall meet paragraph 3.12. of MIL-PRF-28800F for test equipment calibration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

6. PACKAGING.

Preservation, Packaging, Labeling, And Marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order. Each oscilloscope shall have an identification plate permanently affixed in a readable location which includes the identification information contained in paragraph 3.11.2. of MIL-PRF-28800F.

7. NOTES.

7.1. Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified in conversion tables contained in the latest version of Federal Standard No. 376 and all other requirements of this CID are met. If the manufacturer elects to use metric dimensions as the basis upon which to derive or base a mechanical feature, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

7.2. Source of Document Publications. Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Acquisition Printing Service, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094. Phone 215/697-2179.

7.3 Possible Acceptable Products. The following products may meet the salient characteristics of this CID:

Manufacturer	CAGE	Model
LeCroy Corporation, Inc. 700 Chestnut Ridge Road Chestnut, NY 10977-6499	55716	LC374A-CKTRIG
Tektronix, Inc. P. O. Box 500 Beaverton, OR 97077	80009	TDS 754A-1M-5 TDS 754C-1M-5-1R-23

MILITARY INTERESTS:

Custodian:
Air Force-99

Reviewer:
DLA – CC

Preparing Activity:
Air Force-82

Agent:
Air Force-99

Project No. 6625-F071

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.